



**MANIPAL UNIVERSITY
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SEMESTER I

AI6101:3 Probability Distribution

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Sample Spaces and Events

- A **sample space (S)** is the set of all possible outcomes of a random experiment.
- Example: Tossing a coin $\rightarrow S = \{H, T\}$;rolling a die $\rightarrow S = \{1, 2, 3, 4, 5, 6\}$.
- An **event (E)** is a subset of the sample space that represents one or more outcomes.
- **Simple event:** Consists of only one outcome (e.g., getting a 4 in dice).
- **Compound event:** Consists of two or more outcomes (e.g., getting an even number).
- **Certain event:** The whole sample space, probability = 1.

Sample Spaces and Events

- **Impossible event:** The empty set, probability = 0.
- **Mutually exclusive events:** Events that cannot happen together (e.g., odd and even number in one die roll).
- **Exhaustive events:** Set of events covering the whole sample space.
- Probability of an event E :

$$P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total outcomes}}$$

- Visualization is often done using **Venn diagrams** to show relations between events.

Interpretations and Axioms of Probability

- **Classical:** Based on equally likely outcomes.
- **Relative frequency:** Based on experiments.
- **Axiomatic (Kolmogorov):** Modern approach.

Axiom 1: $P(E) \geq 0$.

Axiom 2: $P(S) = 1$.

Axiom 3: If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$.

- Ensures consistency of probability.
- Used in all advanced probability models.
- Important in both theory & real applications.

Addition Rules of Probability

- Used for “either-or” type problems.
- For mutually exclusive events:
 $P(A \cup B) = P(A) + P(B)$
- For general events:
 $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- Prevents double counting.
- Venn diagrams show overlaps clearly.
- Example: Drawing red card or king.
- Very useful in card/dice problems.

Conditional Probability

- Probability of A given B occurred.
- Formula: $P(A | B) = P(A \cap B) / P(B)$
- Requires $P(B) > 0$.
- Used in dependent events.
- Example: Probability of red card if it's a heart = 1.
- Shown using Venn diagrams.
- Also explained with probability trees.
- Foundation for Bayes' theorem.

Multiplication & Total Probability Rules

- Multiplication rule: $P(A \cap B) = P(A)P(B | A)$.
- For independent events: $P(A)P(B)$.
- Total Probability Law:

$$P(A) = \sum P(A|B_i)P(B_i).$$

- Works with partition of sample space.
- Useful for multi-step experiments.
- Helps in medical/engineering problems.

Bayes' Theorem

- Reverses conditional probabilities.
- Example: Medical testing problems.
- Widely used in AI & ML.
- Also applied in spam filtering.
- Very useful in diagnostics.

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum P(A|B_j)P(B_j)}.$$

Bayes' Theorem

- Bayes' Theorem is a fundamental rule in probability that describes how to update the probability of a hypothesis when new evidence is observed.
- It is named after the statistician **Thomas Bayes**.
- The formula is:

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

- Here, $P(A | B)$ is the posterior probability (probability of event A given event B).
- $P(A)$ is the prior probability of A, before considering B.
- $P(B|A)$ is the likelihood, the probability of observing B if A is true.

Bayes' Theorem

- $P(B)$ is the marginal probability of B, acting as a normalizing factor.
- It helps in revising beliefs based on new evidence.
- Common applications include medical diagnosis, spam filtering, and machine learning.
- For example, given a positive medical test result, Bayes' Theorem helps estimate the probability of actually having the disease.
- It bridges prior knowledge with new information for better decision-making.
- Thus, it provides a powerful framework for reasoning under uncertainty.

Random Variables

- Assigns numbers to outcomes.
- **Discrete:** Countable values (e.g., dice).
- **Continuous:** Uncountable (e.g., height).
- Represented by probability distributions.
- Link between probability & statistics.
- Example: Number of heads in 3 tosses.
- Basis of all probability models.

Bernoulli Distribution

- Single trial with two outcomes.
- Success (1) or failure (0).
- PMF:

$$P(X = x) = p^x (1 - p)^{1-x}.$$

- Mean = p .
- Variance = $p(1-p)$.
- Example: Coin toss.
- Foundation for Binomial distribution.

Binomial Distribution

- Repetition of Bernoulli trials.
- n independent trials, success prob = p .

- PMF:

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

- Mean = np .
- Variance = $np(1-p)$.
- Example: Number of heads in 10 tosses.
- Used in quality control & testing.

Poisson Distribution

- Models rare events.
- Used when n large, p small.

- PMF:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

- Mean = λ .
- Variance = λ .
- Example: Calls per hour at call centre.
- Used in traffic, queues, accidents.
- Approximation of Binomial.

Normal Distribution

- Continuous, bell-shaped curve.

- Formula:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

- Mean = Median = Mode.
- Symmetrical about mean.
- 68-95-99.7% rule applies.
- Example: Heights, IQ scores.
- Most used in statistics.
- Basis of hypothesis testing.